Y2 Quantum Mechanics

Part 1 – Wave Mechanics

Observables; operators; wave theory of matter; Schrödinger equation; interpreting the wave-function and expectation values.

Part 2 – Quantum Mechanics in 1D

Solutions of the time-independent Schrödinger equation: potential barriers; quantum tunnelling; square well potentials; simple harmonic oscillator potential.

Part 3 – Principles of Quantum Mechanics

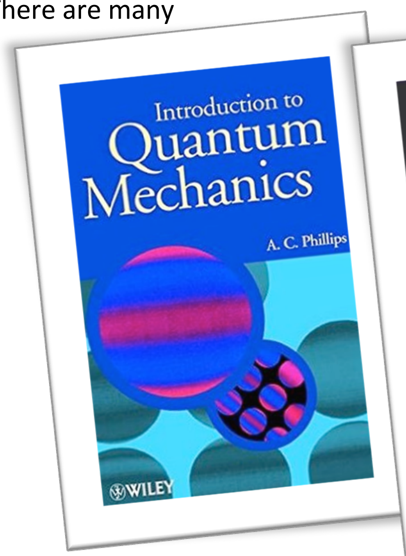
Time dependence; superposition and measurement; simultaneous eigenfunctions; commutation and uncertainty.

Part 4 – Quantum Mechanics in 3D

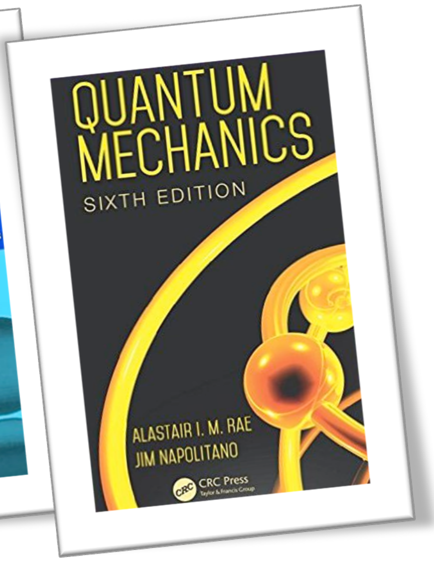
Degeneracy; angular momentum; central potentials; single-electron atoms (hydrogen); single-electron atom wave-functions; Stern-Gerlach experiment – intrinsic spin; total angular momentum; atomic structure; Pauli exclusion principle; Ladder operators.

Textbooks

There are many



Good level for this course



More advanced

See also:

Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles, Eisberg and Resnick

Quantum Physics, S. Gasiorowicz

Quantum Mechanics is a philosophically challenging subject:

However, most were still unhappy with what they had discovered …

'Anyone who is not shocked by quantum theory has not understood it.' **Niels Bohr**
'I think I can safely say that nobody understands quantum mechanics.' **Richard Feynman**
… a some were never reconciled to it

'I am, in fact, firmly convinced that the essentially statistical character of contemporary quantum theory is solely to be ascribed to the fact that this (theory) operates with an incomplete description of physical reality.' **Albert Einstein**

Y1 Quantum Mechanics introduced:

- Blackbody radiation and atomic spectra
- Photoelectric effect
- Compton scattering
- Electron Diffraction
- Photons and electrons can behave both as waves and particles, e.g. from the photoelectric effect and diffraction
- Waves can interfere and superpose
- Energy is proportional to frequency $E = h\nu$
- Momentum is inversely proportional to wavelength $p = h/\lambda$ [de Broglie PhD 1924, Nobel Prize 1929]

Y2 QM will develop the formalism of QM

1. Wave Theory of Matter

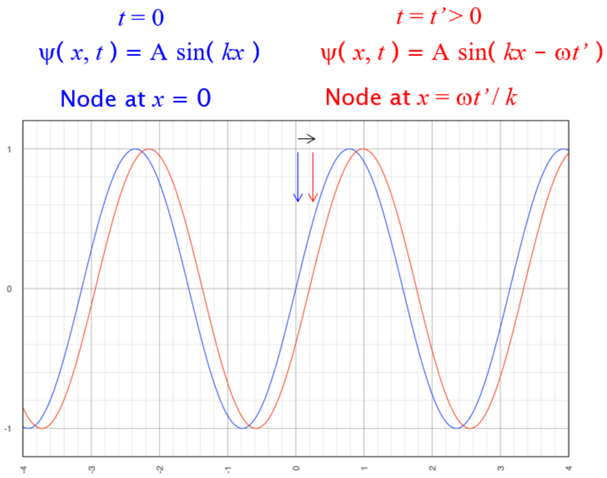
Waves:

Start with a simple sine wave

$\Psi(x,t) = Asin(kx - \omega t)$

Wave propagating in the positive x direction with wavenumber k ($k = 2\pi/\lambda$) and angular frequency ω ($\omega = 2\pi f$).

$(kx - \omega t)$ is the phase of the wave and the velocity at which a point of constant phase moves is $v_{phase} = \omega/k$.



In quantum mechanics we are going to use a complex representation ($z = x + iy$) to describe the waves:

$\Psi(x,t) = Ae^{i(kx-\omega t)} = A[cos(kx - \omega t) + i\ sin(kx - \omega t)]$

How should this complex wave be interpreted?

2. Probability Interpretation

In classical electromagnetism, Maxwell's equations in free space (no charges or currents) predicts that light is described by the oscillation of electric and magnetic fields.

$$\nabla^2 E = \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2}$$

and

$$\nabla^2 B = \frac{1}{c^2}\frac{\partial^2 B}{\partial t^2}$$

Which implies $E = E_0 \sin(kx - \omega t)$ and $B = B_0 \sin(kx - \omega t)$

For waves the power *P* is proportional to the square of the EM fields $P \propto E_0^2$.

In QM light can be thought of as a discrete stream of photons. Each photon has energy $\hbar\omega$

Power in this stream of photons $P = n\hbar\omega$, where n is the number of photons per second per unit area ($m^{-2}s^{-1}$).

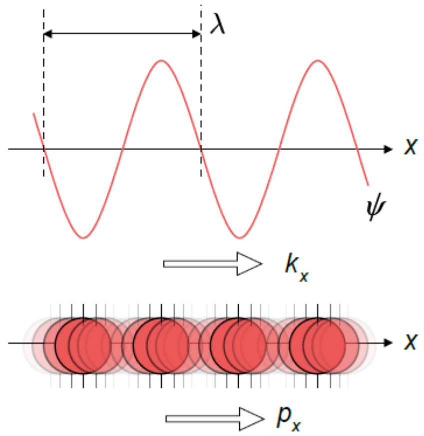
Hence, $n \propto E_0^2$, or in general $n \propto A^2$, where A is the amplitude of the wave, $\Psi(x,t)$.

At low intensity (n<<1) the photons arrive intermittently as discrete flashes.

Wave picture still describes the average power integrated over time – but does not predict the arrival of each photon.

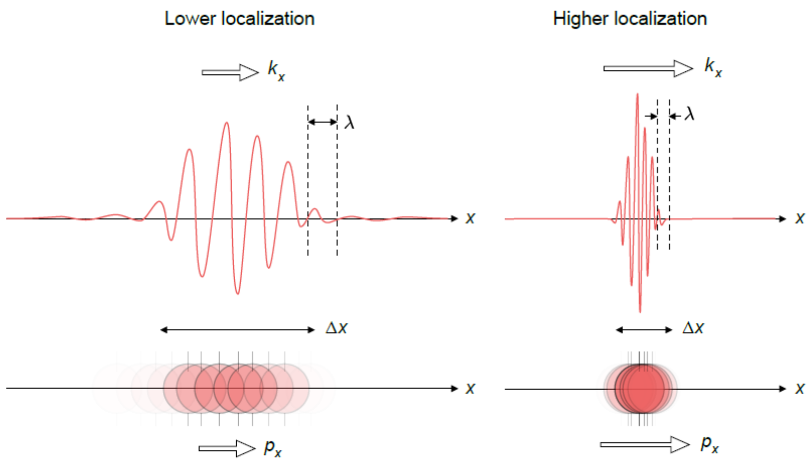
This gives rise to the statistical interpretation – probability of when the photon will arrive.

3. Wave Packets and Uncertainty



Consider the above sine wave. It has well defined direction and momentum but is continuous and hence does not define position.

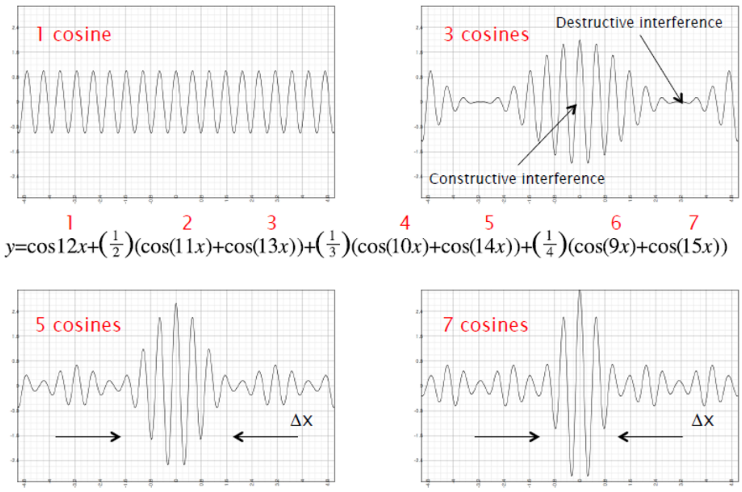
Localisation can be achieved by forming a wave-packet



This is formed from the superposition of many sine waves. For example, the case of adding cosines of the form

$$y(x) = \cos(12x) + \frac{1}{2}\cos(11x) + \frac{1}{2}\cos(13x) + \cos(14x) + \frac{1}{4}\cos(9x) + \frac{1}{4}\cos(15x)$$

Is shown below. As more cosines are added there is greater localisation in space. Each cosine has a different value of k and hence momentum.



Hence as the position become more localised then the momentum becomes more uncertain. This connects to the Heisenberg Uncertainty Principle (HUP)

$$\Delta p . \Delta x \geq \frac{\hbar}{2}$$

[Equality is for Gaussian distribution in x/p, all other distributions give the inequality.]

4. Building the Schrödinger Equation

The wave equation which applies to EM waves or classical mechanical waves has the following form in 1D

$$\frac{\partial^2 \Psi}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 \Psi}{\partial t^2}$$

or in 3D

$$\nabla^2 \Psi = \frac{1}{c^2}\frac{\partial^2 \Psi}{\partial t^2}$$

This can be derived from Maxwell's Equations in EM (see EM2) or in year 1 Waves and Optics.

The QM form of the wave equation cannot be derived in the same way, rather its form is postulated – it cannot be derived from other equations.

Assumptions

1. Consistent with $E = \hbar\omega$ and $p = \hbar k$
2. Satisfies energy conservation:

$$E = T + V = \frac{p^2}{2m} + V(x,t)$$

E is the total energy of a particle of mass m.

For a free particle not bound by a force, $F = -dV/dx = 0$ thus the potential energy is constant $V = V_0$.

Combining conditions 1 and 2 gives

$$\hbar\omega = \frac{\hbar^2 k^2}{2m} + V(x,t)$$

[Note this is non-relativistic. The consideration of relativity leads to the Dirac Equation.]

3. The QM wave equation is a linear differential equation, so superposition applies. This means that if Ψ_1 and Ψ_2 are solutions to the equation then $\Psi = a\Psi_1 + b\Psi_2$ is also a solution.

Hence for a free particle (or equivalently a constant potential), in one dimension, then the solutions to the wave-equation are expected to be sinusoidal – travelling wave:

$$\Psi(x,t) = A[\cos(kx-\omega t) + \gamma\sin(kx-\omega t)],$$

where γ is an unknown constant – which is to be determined.

Calculating the partial derivatives:

$$\frac{\partial\Psi}{\partial t} = A\omega[\sin(kx-\omega t) - \gamma\cos(kx-\omega t)]$$

$$\frac{\partial\Psi}{\partial x} = Ak[-\sin(kx-\omega t) + \gamma\cos(kx-\omega t)]$$

$$\frac{\partial^2\Psi}{\partial x^2} = -Ak^2[\cos(kx-\omega t) + \gamma\sin(kx-\omega t)] = -k^2\psi$$

Comparing this to

$$\hbar\omega = \frac{\hbar^2k^2}{2m} + V_0$$

$$\hbar\omega\Psi = \frac{\hbar^2k^2}{2m}\Psi + V_0\Psi$$

Suggests a differential equation to give the required ω and k dependence:

$$\beta\frac{\partial\Psi}{\partial t} = \alpha\frac{\partial^2\Psi}{\partial x^2} + V_0\Psi$$

Substituting the partial derivatives into this equation

$$\begin{aligned}\beta A\omega[\sin(kx-\omega t) - \gamma\cos(kx-\omega t)]&\\ &= -\alpha Ak^2[\cos(kx-\omega t) + \gamma\sin(kx-\omega t)]\\ &+ V_0A[\cos(kx-\omega t) + \gamma\sin(kx-\omega t)]\end{aligned}$$

Seek a solution that is valid for all x and t.

Gathering sine and cosine terms:

$$(-\alpha k^2 + V_0 + \gamma\beta\omega)\cos(kx-\omega t) + (\gamma(-\alpha k^2 + V_0) - \beta\omega)\sin(kx-\omega t) = 0$$

So need $-\alpha k^2 + V_0 + \gamma\beta\omega = 0$ and $\gamma(-\alpha k^2 + V_0) - \beta\omega = 0$.

So

$$-\alpha k^2 + V_0 = -\gamma\beta\omega \quad [1]$$

and

$$\gamma(-\alpha k^2 + V_0) = \beta\omega \quad [2]$$

Dividing [1] by [2]

$$\frac{1}{\gamma} = -\gamma, i.e. \gamma^2 = -1$$

Hence, $\gamma = \pm i$. Here the sign is arbitrary so choose + i

Thus $\Psi(x,t) = A[\cos(kx-\omega t) + i\sin(kx-\omega t)] = Ae^{i(kx-\omega t)}$

This is the complex wave-function described earlier.

Now compare

$-\alpha k^2 + V_0 = -\gamma\beta\omega$

and

$\hbar\omega = \dfrac{\hbar^2 k^2}{2m} + V_0$

We see that

$$\alpha = -\frac{\hbar^2}{2m} \text{ and } \beta = i\hbar$$

and hence

$$\boxed{-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V_0\psi = i\hbar\frac{\partial\Psi}{\partial t}}$$

This is Schrödinger's equation – a complex 2nd order differential equation. Here $\Psi(x,t)$ is a complex function.

5. Interpretation of the Wave-function $\boldsymbol{\Psi(x,t)}$

Postulate 1: $\Psi(x,t)$ provides a complete description of a quantum state, i.e. it contains all that there is to know about the state.

Probabilities (Probability Density Function)

Let $P(x,t)dx$ be the probability of measuring a particle between x and $x+dx$.

$P(x,t)$ is the Probability Density Function and must be

i) Real
ii) ≥ 0 (everywhere)
iii) Normalised, i.e.
$$\int_{-\infty}^{\infty} P(x,t)dx = 1$$

This implies that the total probability of finding the particle is unity, if the particle exists.

In QM we define $P(x,t) = |\Psi|^2 = \Psi^*\Psi$

This is analogous to the classical wave amplitude, where intensity is $\propto A^2$.

Note: if $\Psi = a + ib$ then $P(x,t) = (a+ib)(a-ib) = a^2 + b^2$

So Ψ is real and ≥ 0 (but still not normalised).

Normalisation

$$\int_{-\infty}^{\infty} \Psi(x,t)^*\Psi(x,t)dx = 1$$

To check the normalisation: if Ψ_1 is a solution of the S.E. then so is $A\Psi_1$ since the S.E. is a linear PDE and the constant A will divide out.

Solve

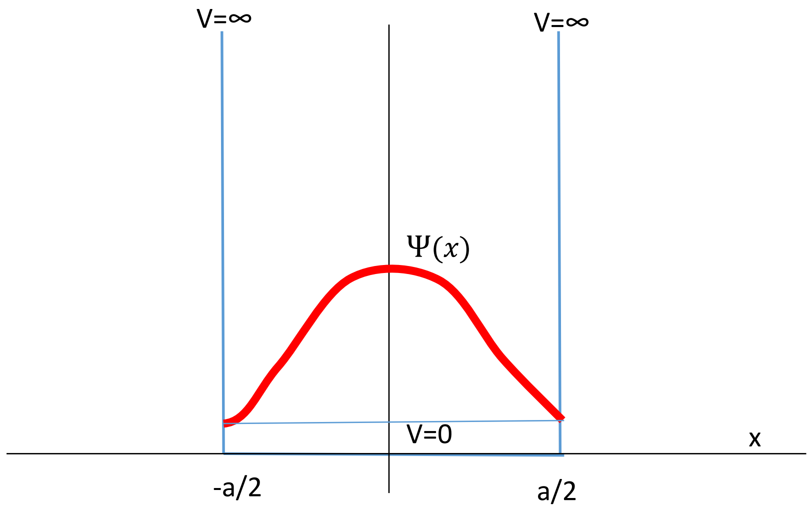
$$\int_{-\infty}^{\infty} A^*\Psi(x,t)^* A\Psi(x,t)dx = |A|^2 \int_{-\infty}^{\infty} \Psi(x,t)^*\Psi(x,t)dx = 1$$

Take an example:

Let

$$\Psi(x,t) = \begin{cases} A\cos\left(\frac{\pi x}{a}\right) e^{-\frac{iEt}{\hbar}} & -\frac{a}{2} \leq x \leq \frac{a}{2} \\ 0 & elsewhere \end{cases}$$

This corresponds to a particle bound in an infinite square potential well.



This is a standing wave.

$$\int_{-\infty}^{\infty} \Psi(x,t)^*\Psi(x,t)dx = 1$$

$$\int_{-\frac{a}{2}}^{\frac{a}{2}} A\cos\left(\frac{\pi x}{a}\right) e^{+\frac{iEt}{\hbar}} A\cos\left(\frac{\pi x}{a}\right) e^{-\frac{iEt}{\hbar}} dx = 1$$

Note the effect of the size of the box:

For a box from -*a* to *a* $A = \sqrt{1/a}$

For a box from -2*a* to 2*a* $A = \sqrt{1/2a}$

$$A^2 \int_{-\frac{a}{2}}^{\frac{a}{2}} \cos^2\left(\frac{\pi x}{a}\right) \quad dx = 1$$

Here we have assumed A is real (otherwise $|A|^2$)

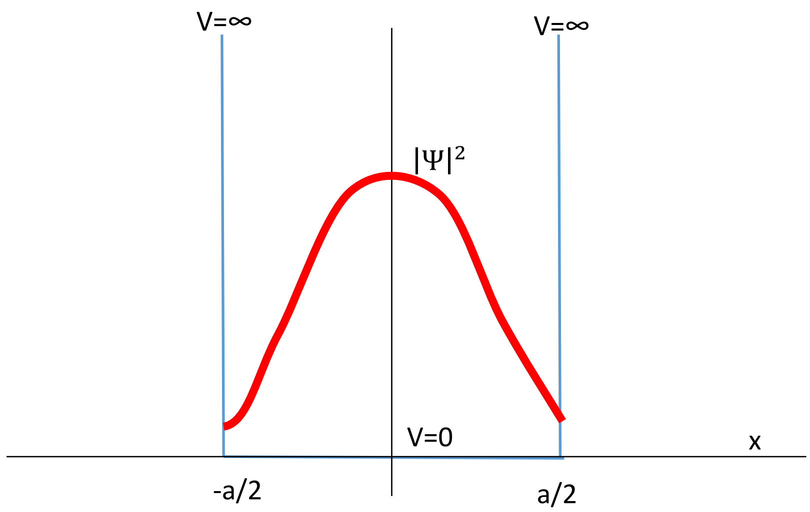
$$\frac{A^2}{2} \int_{-\frac{a}{2}}^{\frac{a}{2}} (1 + \cos\left(\frac{2\pi x}{a}\right)) \, dx = \frac{A^2}{2} \left[x + \underbrace{\frac{a}{2\pi} \sin\left(\frac{2\pi x}{a}\right)}_{=0} \right]_{-a/2}^{a/2} = \frac{A^2}{2} a$$

Hence $A = \sqrt{2/a}$

Note

$$P(x,t) = \Psi^*\Psi = \frac{2}{a}\cos^2\left(\frac{\pi x}{a}\right)$$

is time independent.



Note: probability distribution is not uniform and peaked towards the centre (unlike a classical system).

Expectation Values

Here we will deduce the expectation value for the particles position

$$\langle x \rangle = \int_{-\infty}^{\infty} xP(x,t)dx$$

This is the weighted mean position.

$$\langle x \rangle = \int_{-\infty}^{\infty} \Psi^* x \, \Psi \, dx$$

Where ψ is normalised. For the example before

$$\langle x \rangle = \frac{2}{a}\int_{-\infty}^{\infty} x\cos^2\left(\frac{\pi x}{a}\right) \, dx = 0$$

(since this is an odd function with symmetric limits). So the expectation values gives the expected result!

In general

$$\boxed{\langle O \rangle = \int_{-\infty}^{\infty} \Psi^* \hat{O} \, \Psi \, dx}$$

$\langle O \rangle$ – Observable (e.g. position)

$\hat{O}$ – Mathematical operator

For position $\hat{x} = x$

For momentum $\hat{p} = -i\hbar \, \partial/\partial x$

The momentum operator

A free particle has a well-defined momentum. How does the wave describe this?

$$\Psi = A\, e^{i(kx-\omega t)}$$

$$\frac{\partial \Psi}{\partial x} = A\, ike^{i(kx-\omega t)} = ik\Psi$$

But $p = \hbar k$

$$-i\hbar \frac{\partial \Psi}{\partial x} = p\Psi$$

$$\boxed{\hat{p}\Psi = p\Psi}$$

$\hat{p}$ is the momentum operator: $\hat{p} = -i\hbar\, \partial/\partial x$

p is the momentum eigenvalue (value of the momentum) and is a real number

[eigen: proper, inherent, specific]

Postulate 2: For every physical observable, there is an associated mathematical operator.

If Ψ is the eigenfunction of the particular observable, the result of the operator acting on Ψ is to return the value of the observable multiplied by Ψ.

$$\langle p \rangle = \int_{-\infty}^{\infty} \Psi^* \, \hat{p}\, \Psi \, dx$$

If $\hat{p}\,\Psi = p\,\Psi$ then

$$\langle p \rangle = p \underbrace{\int_{-\infty}^{\infty} \Psi^* \;\; \Psi \, dx}_{=1} = p$$

as expected.

Now consider the particle in the infinite square potential well (a standing wave)

$$\Psi(x,t) = \sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right) e^{-\frac{iEt}{\hbar}}$$

What is the momentum?

$$-i\hbar \frac{\partial}{\partial x}\Psi(x,t) = -i\hbar \frac{\partial}{\partial x}\sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right) e^{-\frac{iEt}{\hbar}} = +\frac{i\hbar\pi}{a}\sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) e^{-\frac{iEt}{\hbar}} \neq p\Psi$$

So Ψ is not an eigenfunction of $\hat{p}$ – why? A standing wave is a superposition of two equal travelling waves, moving in opposite directions.

$$\langle p \rangle = \int_{-a/2}^{a/2} \Psi^* \, \hat{p}\, \Psi \, dx = \int_{-a/2}^{a/2} \sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right) e^{+\frac{iEt}{\hbar}} \frac{i\hbar\pi}{a} \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) e^{-\frac{iEt}{\hbar}} \, dx = 0$$

Since this is an odd function over a symmetric interval.

When $\hat{p}\Psi \neq p\Psi$, $\langle p \rangle$ corresponds to the average value of momentum after many measurements of a particle in the same quantum state.

Note the order of operations matters, e.g. in general:

$$\Psi^*\left(\frac{\partial}{\partial x}\Psi\right) \neq \left(\frac{\partial \Psi^*}{\partial x}\right)\Psi \neq \frac{\partial}{\partial x}(\Psi^*\Psi)$$

Kinetic Energy

$$\hat{T} = \frac{\hat{p}^2}{2m} = \frac{1}{2m}\left(-i\hbar\frac{\partial}{\partial x}\right)\left(-i\hbar\frac{\partial}{\partial x}\right) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$$

6. The Time Independent Schrodinger Equation, TISE

If the potential is independent of time $V(x,t) = V(x)$:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V(x)\Psi = i\hbar\frac{\partial\Psi}{\partial t}$$

LHS depends on x, RHS depends on t. Solve using the technique of separation of variables

Let $\Psi(x,t) = \psi(x)\phi(t)$

$$\left(-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2} + V(x)\psi(x)\right)\phi(t) = \left(i\hbar\frac{\partial\phi(t)}{\partial t}\right)\psi(x)$$

Divide by $\Psi(x,t) = \psi(x)\phi(t)$

$$\frac{\left(-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x)\right)}{\psi(x)} = \left(\frac{i\hbar}{\phi(t)}\frac{d\phi(t)}{dt}\right) = a$$

Where a is a constant. Now have 2 ODEs (note partial derivatives not required any more).

Consider the time-dependence first.

$$i\hbar\frac{d\phi(t)}{dt} = a\phi(t)$$

$$\int_{\phi(0)}^{\phi(t)}\frac{d\phi(t)}{\phi(t)} = -\frac{ai}{\hbar}\int_0^t dt$$

$$\phi(t) = \phi(0)e^{-\frac{iat}{\hbar}}$$

If we set $a = \hbar\omega = E$, then

$$\phi(t) = \phi(0)e^{-i\omega t} = \phi(0)e^{-\frac{iEt}{\hbar}}$$

Note this is the time dependent form of our complex wave. Hence

$$\frac{\left(-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x)\right)}{\psi(x)} = E$$

Or

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

This is the time independent SE. Note this equation is real valued, i.e. E is real and ψ is real, but Ψ is still complex.

This equation has the form

$$(\hat{T} + \hat{V})\psi = E\psi$$

Where $\hat{T}$ and $\hat{V}$ are operators associated with KE and PE and E is the eigen-energy (total energy) of the particle in state Ψ.

This can be replaced by

$$\hat{H}\,\psi = E\psi$$

Where $\hat{H}$ is called the Hamiltonian and is associated with the total energy operator.

7. Free Particle Solutions of the Time Independent Schrodinger Equation [TISE]: Barriers

Take the TISE for a free particle where $V = V_0$. $KE = T = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V_0\psi(x) = E\psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} = -\frac{2m}{\hbar^2}(E - V_0)\psi(x) = -k^2\psi(x)$$

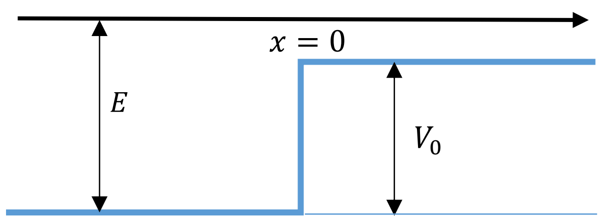
This has the form of SHM, so the solution is oscillatory

$$\psi = \underbrace{A\cos(kx) + B\sin(kx)}_{standing\ wave\ representation} = \underbrace{Ce^{ikx} + De^{-ikx}}_{travelling\ wave\ representation}$$

With

$$k = \frac{1}{\hbar}\sqrt{2m(E - V_0)}$$

Potential Step



$$V(x) = \begin{cases} 0, & x \leq 0 \\ V_0, & x > 0 \end{cases}$$

Classically, the particle will cross the step since $E > V_0$. What happens in QM?

Note such step potentials are not realistic

$$F = -\frac{dV}{dx} = \infty$$

But since $dt \to 0$, the impulse $I = \int F.dt$ is finite. Nevertheless, steps allow the study of some QM principles.

Approach here is to solve the TISE for $x > 0$ and $p = \hbar k$.

We expect free-particle solutions in both regions.

$\Psi(x,t) = Ae^{i(kx-\omega t)} = Ae^{ikx}e^{-i\omega t}$

From the properties of the free particle solution of the TISE let's take the travelling wave form

$$\psi = \underbrace{Ae^{ikx}}_{wave\ travels\ L\ to\ R:\ p=\hbar k} + \underbrace{Be^{-ikx}}_{wave\ travels\ R\ to\ L:\ p=-\hbar k}$$

With

$$k = \frac{1}{\hbar}\sqrt{2m(E-V_0)}$$

Region I $x \leq 0$: $V = 0$

$\psi_1 = Ae^{ik_1x} + Be^{-ik_1x}$

$$k_1 = \frac{1}{\hbar}\sqrt{2mE}$$

Region II $x > 0$: $V = V_0$

$\psi_2 = Ce^{ik_2x} + De^{-ik_2x}$

$$k_2 = \frac{1}{\hbar}\sqrt{2m(E-V_0)}$$

Note $k_1 > k_2$ and hence $p_1 > p_2$

Need to apply the boundary conditions to deduce the values of A, B, C & D.

i) In Region II ($x > 0$), there is no source of reflected wave so $D = 0$. There is only a possibility of a reflection if the wave encounters a change in potential.

ii) At the boundary ($x = 0$), the wave-function must be continuous otherwise the value of $\frac{d\psi}{dx} = \infty$ (i.e. momentum is infinite).

iii) At the boundary ($x = 0$), the derivative of the wave-function wrt x must be continuous otherwise the value of $\frac{d^2\psi}{dx^2} = \infty$ (i.e. kinetic energy is infinite).

From ii)

$\psi_1(0) = \psi_2(0)$

$Ae^{ik_10} + Be^{-ik_10} = Ce^{ik_20}$

$A + B = C \qquad$ (**)

From iii)

$\psi_1{}'(0) = \psi_2{}'(0)$

$ik_1(Ae^{ik_10} - Be^{-ik_10}) = ik_2Ce^{ik_20}$

$A - B = Ck_2/k_1 \qquad$ (**)

Note that cannot have $B = 0$, unless $k_1 = k_2$.

Find B and C in terms of the amplitude of the original wave amplitude A.

Adding the two equations **

$2A = C(1 + k_2/k_1)$

$C = \frac{2Ak_1}{k_1 + k_2}$

Subtracting **

$2B = C(1 - k_2/k_1)$

$B = A\frac{k_1 - k_2}{2k_1}\frac{2k_1}{k_1 + k_2}$

$B = A\frac{k_1 - k_2}{k_1 + k_2}$

B is the reflected amplitude and C is the transmitted amplitude. Similar to waves on a string or waves refracted by optical medium.

Transmission Probability

Probability of transmission T = transmitted flux/incident flux

Flux= number density x velocity

In 3d: flux $[T]^{-1}[L]^{-2}$ = $[L]^{-3}$ x $[L][T]^{-1}$

In 1d: flux $[T]^{-1}$ = $[L]^{-1}$ x $[L][T]^{-1}$

i.e. measure the number of particles passing a particular point on the x-axis per unit time.

Number density is proportional to ψ^2 and velocity is proportional to k, so incident flux is proportional to $|A|^2 k_1$.

Hence,

$$T = \frac{k_2|C|^2}{k_1|A|^2} = \frac{k_2}{k_1}\left|\frac{C}{A}\right|^2 = \frac{4k_1k_2}{(k_1 + k_2)^2}$$

Reflection Probability

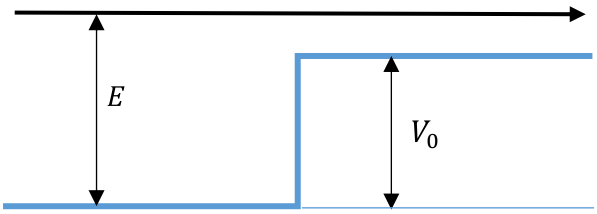
Probability of reflection R = reflected flux/incident flux

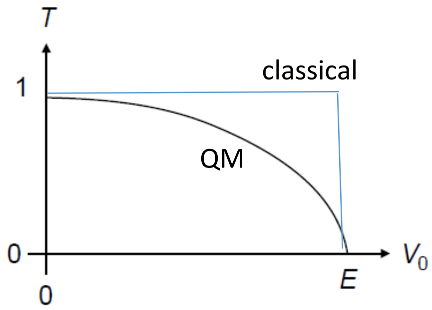
$$R = \frac{k_1|B|^2}{k_1|A|^2} = \left|\frac{B}{A}\right|^2 = \frac{(k_1 - k_2)^2}{(k_1 + k_2)^2}$$

$$T + R = \frac{4k_1k_2}{(k_1 + k_2)^2} + \frac{(k_1 - k_2)^2}{(k_1 + k_2)^2} = \frac{k_1^2 + k_2^2 - 2k_1k_2 + 4k_1k_2}{(k_1 + k_2)^2} = 1$$

Conservation of flux.

How does the transmission vary with the height of the step?

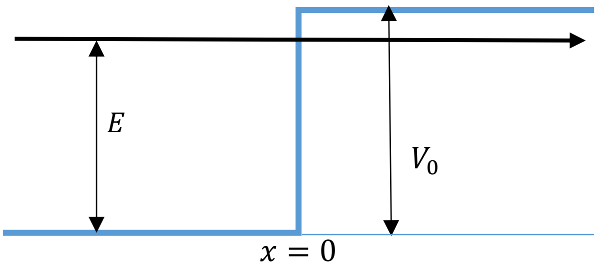




If $V_0 = 0$, then $k_2 = k_1$ and $T = 1, R = 0$

If $V_0 = E$, then $k_2 = 0$ and $T = 0, R = 1$

Potential Step with $V_0 > E$



Classically the particle cannot penetrate the step, so is fully reflected.

What about in QM? Solve the boundary problem

$x \leq 0$: $\psi_1(x) = Ae^{ik_1x} + Be^{-ik_1x}$; $k_1 = \sqrt{2mE}/\hbar$

This is as before.

$x > 0$: $\psi_2(x) = Ce^{ik_2x} + De^{-ik_2x}$; $k_2 = \sqrt{2m(E - V_0)}/\hbar$

However since the energy is less than the height of the potential then the wavenumber becomes imaginary

$$k_2 = \frac{i\sqrt{2m(V_0 - E)}}{\hbar} = ik$$

$$\psi_2(x) = Ce^{-kx} + De^{kx}$$

Now the solutions are not oscillatory, but exponential.

Applying the boundary conditions:

i) As $x \to \infty$, $e^{kx} \to \infty$ so this implies $D = 0$.
ii) $\psi(x)$ is continuous at $x = 0$, $A + B = C$ **
iii) $\psi'(x)$ is continuous at $x = 0$, $ik_1(A - B) = -kC$
$A - B = ik/k_1 C$ **

Using the two ** equations.

$$2A = C\left(1 + \frac{ik}{k_1}\right)$$

$$2B = C\left(1 - \frac{ik}{k_1}\right)$$

$$\frac{B}{A} = \frac{k_1 - ik}{k_1 + ik}$$

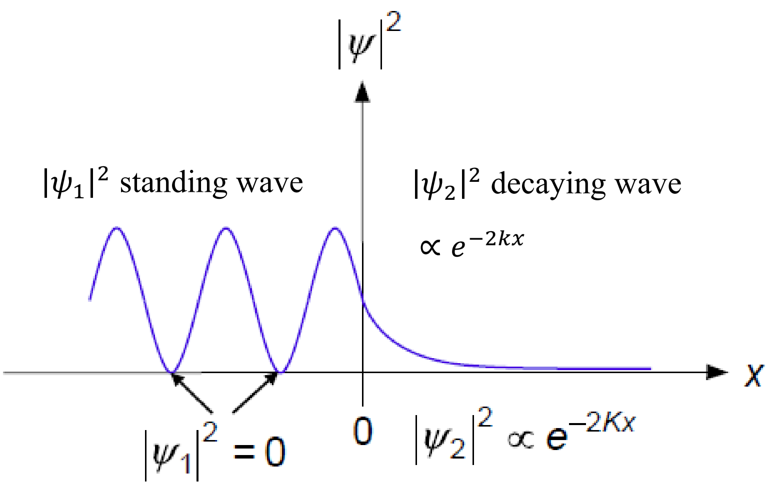
Which is complex.

Reflection probability:

$$R = \left|\frac{B}{A}\right|^2 = \left|\frac{k_1 - ik}{k_1 + ik}\right|^2$$

$$R = \left|\frac{B}{A}\right|^2 = \frac{B^*B}{A^*A} = \frac{k_1 + ik}{k_1 - ik}\,\frac{k_1 - ik}{k_1 + ik} = 1$$

Same as classical result, but with an important difference. Note T now becomes complex and has no physical meaning.



For the region in front of the barrier there is a standing wave, where the incident and reflected waves interfere and the PDF is not uniform.

For the region inside of the barrier there is a decaying wave, with a finite probability of finding the particle inside the barrier.

i) **For $x \le 0$ $\psi_1(x) = Ae^{ik_1x} + Be^{-ik_1x}$**

Sub for amplitudes

$$\psi_1(x) = \frac{C}{2}\left(1 + \frac{ik}{k_1}\right)e^{ik_1x} + \frac{C}{2}\left(1 - \frac{ik}{k_1}\right)e^{-ik_1x}$$

$$\psi_1(x) = C\left\{\frac{e^{ik_1x} + e^{-ik_1x}}{2} + \frac{ik}{k_1}\frac{e^{ik_1x} - e^{-ik_1x}}{2}\right\}$$

$$\psi_1(x) = C\left\{\cos(k_1x) - \frac{k}{k_1}\sin(k_1x)\right\}$$

$$\frac{k}{k_1} = \sqrt{\frac{V_0 - E}{E}}$$

This represents a standing wave with nodes when $\psi_1(x) = 0$

$$\cos(k_1 x) = \frac{k}{k_1}\sin(k_1 x)$$

$$x = \frac{1}{k_1}\tan^{-1}(\frac{k_1}{k})$$

Reflecting the softness of the barrier.

ii) For $\boldsymbol{x > 0}$ $\boldsymbol{\psi_2(x) = Ce^{-kx}}$

What can learn about the particle for this classically forbidden region?

Let's estimate the mean distance the particle travels into the forbidden region.

$$\langle x \rangle = \frac{\int_0^\infty \psi^* x \psi dx}{\int_0^\infty \psi^* \psi dx}$$

Note $\int_0^\infty A^*\psi^* A\psi dx = 1$ for normalisation and thus $A^2 = \frac{1}{\int_0^\infty \psi^*\psi dx}$ $\langle x \rangle = \int_0^\infty A^*\psi^* x A\psi dx = \frac{\int_0^\infty \psi^* x\psi dx}{\int_0^\infty \psi^*\psi dx}$

Where the normalisation of the wave-function is accounted for.

$$\langle x \rangle = \frac{\int_0^\infty x e^{-2kx} dx}{\int_0^\infty e^{-2kx} dx}$$

$$\int_0^\infty e^{-2kx} dx = \frac{-1}{2k}[e^{-2kx}]_0^\infty = \frac{1}{2k}$$

$$\int_0^\infty \underbrace{x}_{u} \underbrace{e^{-2kx}}_{dv} dx = \underbrace{\left[\frac{x}{-2k}e^{-2kx}\right]_0^\infty}_{uv=0} + \underbrace{\int_0^\infty \frac{e^{-2kx}}{2k} dx}_{-\int vdu} = \left[\frac{1}{2k}\right]^2$$

$$\langle x \rangle = \frac{\int_0^\infty x e^{-2kx} dx}{\int_0^\infty e^{-2kx} dx} = \frac{1}{2k}$$

Let the uncertainty in $x, \Delta x$, be $\langle x \rangle$

Then if we use the HUP $\Delta x \Delta p \geq \hbar/2$

$\Delta p \geq \hbar k$

$\Delta p \geq \sqrt{2m(V_0 - E)}$

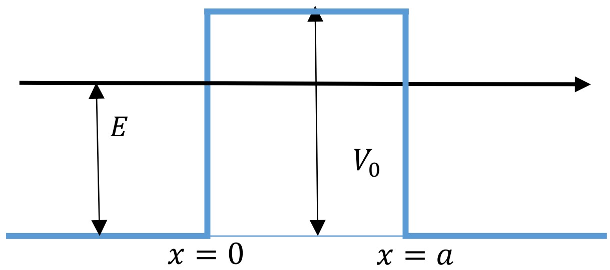
$$\Delta E = \frac{\Delta p^2}{2m} \geq V_0 - E$$

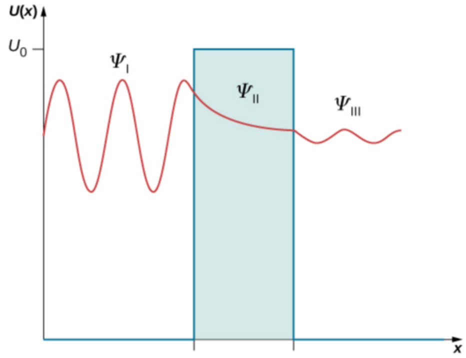
$$\Delta E \geq V_0 - E$$

This implies that if it is possible to constrain the location of the particle to a precision $\langle x \rangle$ then this means that the energy could be larger than the height of the step – not possible as then could have transmission. Hence, the uncertainty on energy constrains our knowledge of where the particle is inside the barrier.

Conceptually analogous to evanescent waves in EM2: in both cases, particle motion cannot be discontinuous at the boundary.

8. Quantum Tunnelling





$$V(x) = \begin{cases} 0, & x \leq 0 \\ V_0, & 0 < x < a \\ 0, & x \geq a \end{cases}$$

Here the particle would be classically reflected.

We now have 3 regions and 2 boundaries.

$x \leq 0$: $\psi_1 = Ae^{ik_1x} + Be^{-ik_1x}$

$0 < x < a$: $\psi_2 = Ce^{-kx} + De^{kx}$

$x \geq a$: $\psi_3 = Fe^{ik_1x}$ [no reflected wave]

Here we have not set $D = 0$ as $x \nrightarrow \infty$.

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

$$k = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

Assert boundary conditions at

$x = 0$: $\psi_1(0) = \psi_2(0)$ and $\psi_1{}'(0) = \psi_2{}'(0)$

$x = a$: $\psi_2(a) = \psi_3(a)$ and $\psi_2{}'(a) = \psi_3{}'(a)$

We would like to find the tunnelling probability

$$T = \frac{k_1}{k_1}\left|\frac{F}{A}\right|^2$$

So we need to relate the transmitted amplitude F to the incident amplitude A.

Procedure:

i) Find A in terms of C and D using boundary conditions at $x = 0$.
ii) Find C and D in terms of F using the boundary conditions at $x = a$

The various amplitudes are related by the requirement that ψ and ψ' must be continuous across the two boundaries.

$x = 0$: Continuity of ψ $\Rightarrow$ $A + B = C + D$ (1)

Continuity of ψ' $\Rightarrow$ $ik_1(A - B) = -k(C - D)$

$$\Rightarrow \quad A - B = \frac{ik}{k_1}(C - D) \qquad (2)$$

$$x = a\text{: Continuity of } \psi \qquad \Rightarrow \quad Ce^{-ka} + De^{ka} = Fe^{ik_1 a} \qquad (3)$$

$$\text{Continuity of } \psi' \qquad \Rightarrow \quad -k(Ce^{-ka} - De^{ka}) = ik_1 Fe^{ik_1 a}$$

$$\Rightarrow \quad \frac{-ik_1}{k} Fe^{ik_1 a} = (Ce^{-ka} - De^{ka}) \qquad (4)$$

The conditions at $x = 0$ allow us to write A in terms of C and D. Adding (1) and (2) we obtain:

$$A = \frac{C}{2k_1}(k_1 + ik) + \frac{D}{2k_1}(k_1 - ik) \qquad (5)$$

The conditions at $x = a$ meanwhile allow us to write C and D in terms of F. By adding/subtracting (4) to/from (3) we obtain:

$$D = \frac{1}{2} Fe^{ik_1 a} e^{-ka} \left(1 + \frac{ik_1}{k}\right) = \frac{F}{2k}(k + ik_1) e^{ik_1 a} e^{-ka}$$

$$C = \frac{1}{2} Fe^{ik_1 a} e^{ka} \left(1 - \frac{ik_1}{k}\right) = \frac{F}{2k}(k - ik_1) e^{ik_1 a} e^{ka}$$

and substituting these into (5) gives:

$$A = \frac{F}{4kk_1}(k - ik_1)(k_1 + ik) e^{ik_1 a} e^{ka} + \frac{F}{4kk_1}(k + ik_1)(k_1 - ik) e^{ik_1 a} e^{-ka}$$

$$A = \frac{F}{4kk_1}(i(k - ik_1)^2 e^{ka} - i(k + ik_1)^2 e^{-ka}) e^{ik_1 a}$$

$$A = \frac{iF}{4kk_1}((k - ik_1)^2 e^{ka} - (k + ik_1)^2 e^{-ka}) e^{ik_1 a}$$

Let's take the result:

$$A = \frac{iF}{4kk_1}[(k - ik_1)^2 e^{ka} - (k + ik_1)^2 e^{-ka}] e^{ik_1 a}$$

For a high barrier k is large

For a wide barrier a is large

For a high wide barrier $ka \gg 1,\ e^{-ka} \to 0$

Hence

$$A = \frac{iF}{4kk_1}[(k - ik_1)^2 e^{ka}] e^{ik_1 a}$$

$$\frac{F}{A} = \frac{4kk_1}{i(k - ik_1)^2} e^{-ka} e^{-ik_1 a}$$

$$\frac{F^*}{A^*} = \frac{4kk_1}{-i(k - ik_1)^2} e^{-ka} e^{ik_1 a}$$

and

$$T = \left|\frac{F}{A}\right|^2 = \frac{F^* F}{A^* A} = \frac{16k_1^2 k^2}{(k^2 + k_1^2)^2} e^{-2ka}$$

$$k = \frac{\sqrt{2m(V_0 - E)}}{\hbar}; \quad k_1 = \frac{\sqrt{2mE}}{\hbar}$$

$$\frac{16k_1^2 k^2}{(k^2 + k_1^2)^2} = \frac{16(V_0 - E)E}{(V_0 - E + E)^2} = \frac{16E}{V_0}\left(1 - \frac{E}{V_0}\right)$$

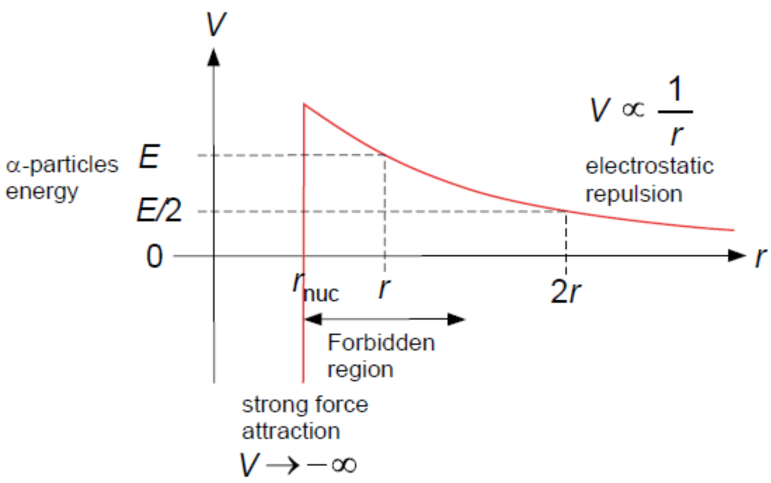
$$T = \frac{16E}{V_0}\left(1 - \frac{E}{V_0}\right) e^{-2ka}$$

For regions

$x < 0$ we have standing waves, but as reflection is not 100% don't get complete destructive interference.

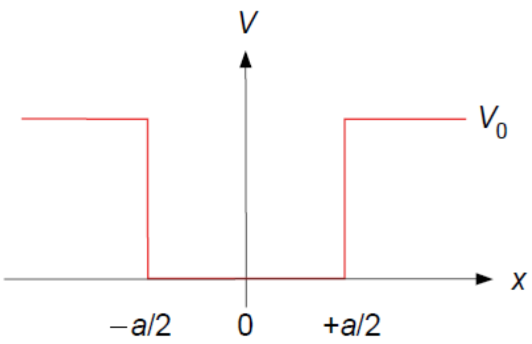
$x > a$ we have travelling waves, with amplitude F

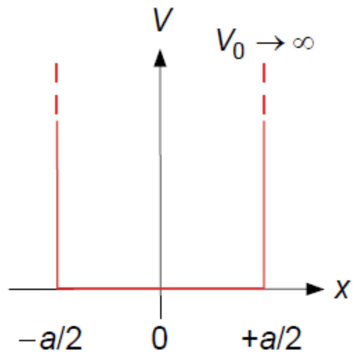
Example of Alpha-decay



- Exponential dependence of the probability for alpha-decay on the barrier height and thickness

9. Square Well Potential





Infinite Well

Here we are going to explore solutions of the TISE for square well potentials. We will start with the simplest form, the infinite potential well (right image above).

- Outside the well $|x| \geq \frac{a}{2}, V(x) = \infty$ and hence $\psi(x) = 0$ $[\psi_2(x) \propto e^{-kx}]$
- Inside the well $|x| < \frac{a}{2}, V(x) = 0$

Inside: $\psi(x) = A'e^{ikx} + B'e^{-ikx}$ or $\psi(x) = Asin(kx) + Bcos(kx)$, both satisfy the TISE, and one can be rearranged to be the other $\left[e^{i\theta} = cos\theta + isin\theta, e^{-i\theta} = cos\theta - isin\theta\right]$. For simplicity we will use the trigonometric form.

Again we will apply the boundary conditions such that $\psi\left(-\frac{a}{2}\right) = \psi\left(\frac{a}{2}\right) = 0$.

$$x = -\frac{a}{2}: \; -Asin\left(\frac{ka}{2}\right) + Bcos\left(\frac{ka}{2}\right) = 0 \quad **$$

$$x = +\frac{a}{2}: \; +Asin\left(\frac{ka}{2}\right) + Bcos\left(\frac{ka}{2}\right) = 0 \quad **$$

Note that ψ' is not continuous at the boundaries since $V = \infty$ is not physical, and potential is not realistic from a real world perspective.

Adding and subtracting ** equations

$$2Bcos\left(\frac{ka}{2}\right) = 0$$

$$2Asin\left(\frac{ka}{2}\right) = 0$$

Cannot satisfy these equations simultaneously for the same value of $ka/2$.

Either: $A = 0$ and $\cos\left(\frac{ka}{2}\right) = 0$

Or: $B = 0$ and $\sin\left(\frac{ka}{2}\right) = 0$

So have either sine or cosine solutions, but cannot have mixtures of sines and cosines.

$\cos\,(ka/2) = 0$ when $k = \pi/a, 3\,\pi/a, 5\,\pi/a$

$\sin\,(ka/2) = 0$ when $k = 0, 2\,\pi/a, 4\,\pi/a$

But $k = 0$ is not a solution as $\psi(x) = Asin(0.x)$ for all values of x does not describe a particle in a well as it is zero everywhere.

We thus have two sets of solutions

i)	$\psi_n(x) = B_n \cos(k_n x)\,;\, k_n = n\pi/a \;\; (n = 1,3,5,7 \ldots \ldots)$
ii)	$\psi_n(x) = A_n \sin(k_n x)\,;\, k_n = n\pi/a \;\; (n = 2,4,6,8 \ldots \ldots)$

If we normalise these wavefunctions we find that $A_n = B_n = \sqrt{2/a}$ for all solutions.

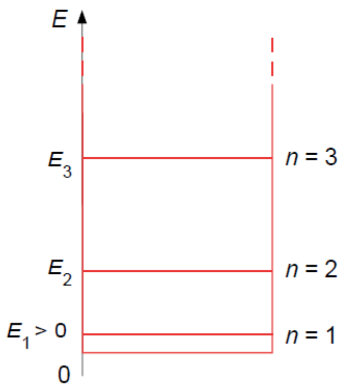
The associated energy eigenvalues are

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

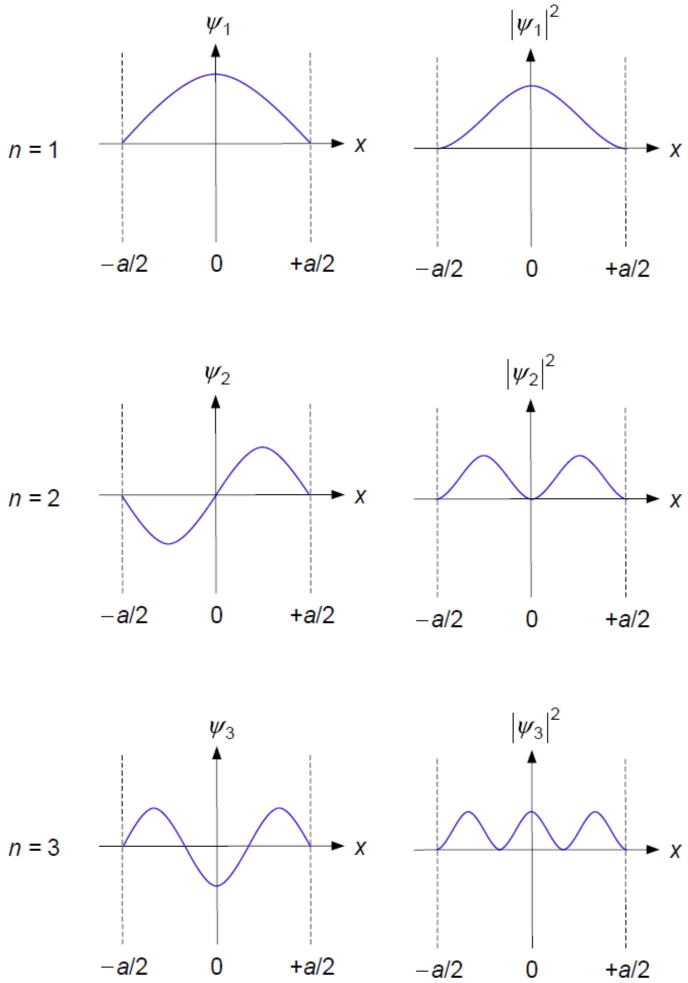
Energy of bound states are quantised, only discrete values of E_n allowed.

n is a quantum number that labels the solutions of the TISE-equation.

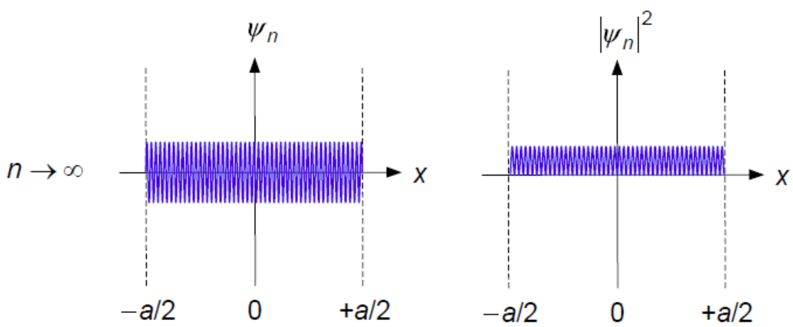
Note that $n = 0$ is not a solution and hence the lowest energy state is not at zero energy (called zero point energy). [HUP $\Delta x < \infty$ hence $\Delta p > 0$ and hence $\Delta E > 0$ and thus the energy of the state is not zero.]



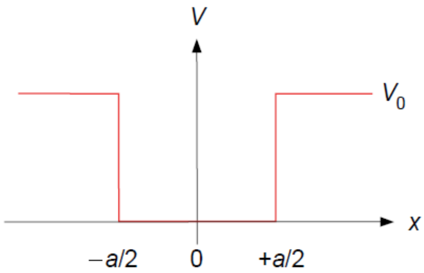
Solutions:



- Probability density is not uniform across the well
- Two sets of solutions have different parities (even and odd): $\psi_1$and ψ_3 are even (reflection symmetric) and $\psi_2$is odd (reflections asymmetric).
- In the high energy limit (as $n \to \infty$) the QM solutions approach classical solutions (spacing between oscillations becomes negligible and distribution become continuous); the **correspondence principle** states that in the limit of large quantum numbers the predictions of quantum physics become identical to the predictions of classical physics**.**



Finite Well



$$V(x) = \begin{cases} V_0, & x \le -\frac{a}{2} \\ 0, & -\frac{a}{2} < x < \frac{a}{2} \\ V_0, & x \ge \frac{a}{2} \end{cases}$$

We need, again, to consider the solutions for the three regions

$$x \le -a/2:\ \psi_1(x) = Ae^{Kx} + \underbrace{Be^{-Kx}}_{ignore\ as\ diverges} \quad ; \ K = \sqrt{2m(V_0 - E)}/\hbar$$

$$|x| < a/2:\ \psi_2(x) = C\sin(kx) + D\cos(kx); \ \ k = \sqrt{2mE}/\hbar$$

$$x \ge +a/2:\ \psi_3(x) = Fe^{-Kx} + \underbrace{Ge^{Kx}}_{ignore\ as\ diverges} \quad ; \ K = \sqrt{2m(V_0 - E)}/\hbar$$

Here we will not be rigorous in the derivation of the solution (the details are provided on Canvas).

Approach is:

Apply boundary conditions and this time $\psi(x)$ and $\psi'(x)$ are both continuous at $x = \pm a/2$.

Find that

i) Either $C = 0$ and $A = F$, or $\cot{(ka/2)} = -K/k$
ii) Either $D = 0$ and $A = -F$, or $\tan{(ka/2)} = +K/k$

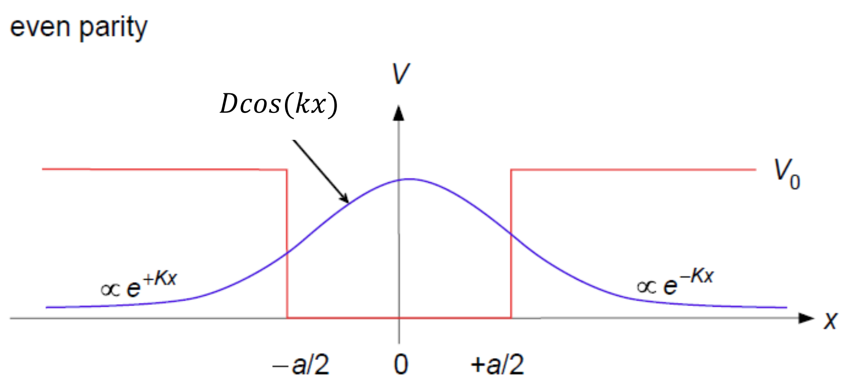
One of i) must be satisfied simultaneously with one of ii), which gives rise to 4 possibilities.

1) If $C = D = 0$ then $\psi_2(x) = 0$; this is not a valid solution
2) If $\cot{(ka/2)} = -K/k$ and $\tan{(ka/2)} = +K/k$ then $\tan^2(ka/2) = -1$; this is not a valid solution as k and a are both real.

So we end up with two valid solutions

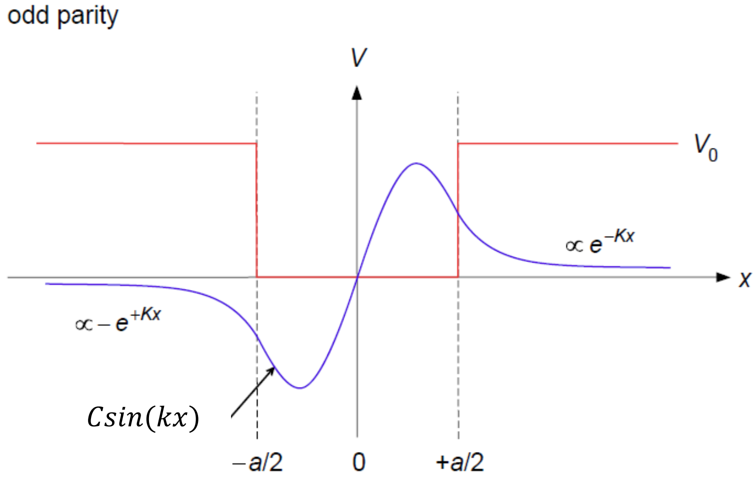
3) $C = 0$ and $A = F$ and $\tan{(ka/2)} = +K/k$

$\psi_2(x) = Dcos(kx)$ and $\psi_1(-x) = \psi_3(+x)$; even parity



4) $D = 0$ and $A = -F$ and $\cot{(ka/2)} = -K/k$

$\psi_2(x) = Csin(kx)$ and $\psi_1(-x) = -\psi_3(+x)$; odd parity



Note that the wavelength of the standing waves has increased wrt the infinite well. The longer wavelength implies smaller wavenumber (momentum) and hence lower energy. So the energy level will be lower for a potential well of the same width, compared with the infinite potential well.

To find the energy eigenvalues, E_n, then need to solve for k.

Even parity:

$$\tan\left(\frac{ka}{2}\right) = +\frac{K}{k}$$

Odd parity:

$$\cot\left(\frac{ka}{2}\right) = -\frac{K}{k}$$

These are called transcendental equations. These cannot be solved analytically; they need to be solved either numerically or graphically.

Substitute for wavenumbers

$$\tan\left(\frac{a}{\hbar}\sqrt{\frac{mE}{2}}\right) = +\sqrt{\frac{V_0 - E}{E}}$$

$$-\cot\left(\frac{a}{\hbar}\sqrt{\frac{mE}{2}}\right) = +\sqrt{\frac{V_0 - E}{E}}$$

If we set

$$y = \frac{a}{\hbar}\sqrt{\frac{mE}{2}}$$

Then

$$E = \frac{2\hbar^2}{ma^2}y^2$$

$$\tan(y) = \sqrt{\frac{V_0 - \frac{2\hbar^2}{ma^2}y^2}{\frac{2\hbar^2}{ma^2}y^2}} = \sqrt{\frac{\frac{ma^2}{2\hbar^2}V_0 - y^2}{y^2}}$$

Set the set of constant terms to another constant:

$$\lambda = \frac{ma^2}{2\hbar^2}V_0$$

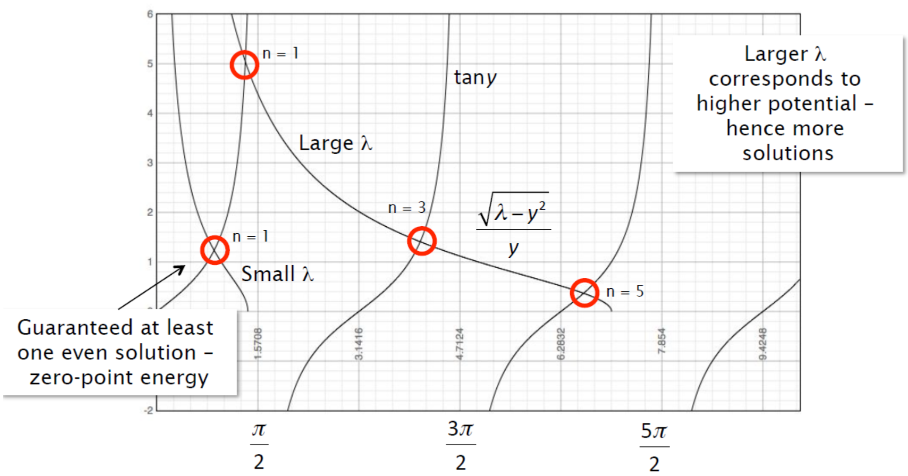
The transcendental equations then become, for even and odd parity respectively:

$$\tan(y) = +\frac{\sqrt{\lambda - y^2}}{y}$$

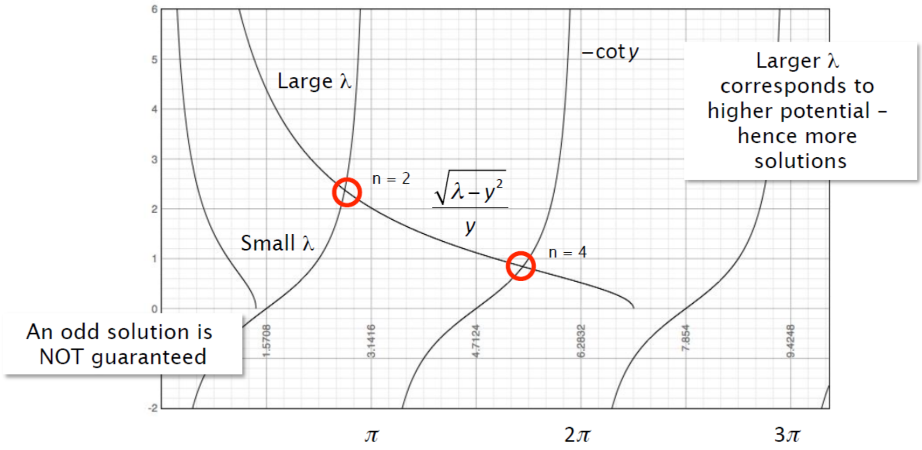
$$\cot(y) = -\frac{\sqrt{\lambda - y^2}}{y}$$

Plot LHS and RHS of functions on same graph and intersections give solutions for y and hence E.

Even solutions

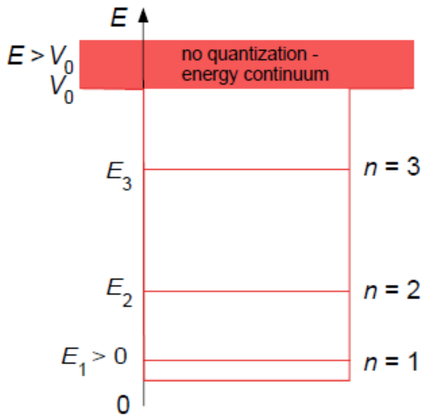


Odd solutions



Notes:

- As $\lambda \to \infty$ $(V_0 \to \infty)$ then the solutions move closer to the poles of the tan and cot functions, i.e. closer to the infinite square well.
- For finite λ, solutions lie at lower values of y and hence E.
- As $E \to V_0, y^2 \to \lambda$ then the RHS tends to zero and the higher intersections are lost (finite number of eigenvalues).
- Provided $V_0 > 0$ $(\lambda > 0)$ then there is always at least one positive (even) parity solution.



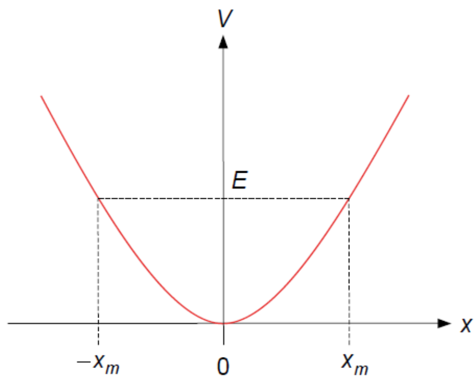
10. The Simple Harmonic Oscillator

The SHO is much more realistic because there are no abrupt changes in the potential as with the square well. It is interesting also because several quantum states behave as oscillators.

SHM arises when

$F = m\ddot{x} = -kx$

$\ddot{x} = -\frac{k}{m}x$



With solutions

$x(t) = A\sin(\omega t) + B\cos(\omega t)$

$\omega = \sqrt{\frac{k}{m}}$

Potential:

$F = -\frac{dV}{dx};\ V(x) = \int_0^x kx'dx' = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2$

Classically any energy is allowed, but in QM only certain eigenvalues are permitted.

Solve the TISE for the above potential. Since the potential is not constant, free particle solutions will not be valid ($k \propto \sqrt{E-V}$) . Wavenumber k varies across the well. This can be understood as $E = T + V = constant$ for a state and hence there is an exchange between KE and PE.

Solution gives

- Infinite number of discrete eigenvalues
- Eigenfunctions are similar to square-well and alternate in terms of odd and even parity
- Eigenfunctions decay into the classically forbidden region beyond the potential.

The TISE is:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2x^2\psi = E\psi$$

With $u = x\sqrt{(m\omega/\hbar)}$

The solution for the eigenfunctions is (for derivation see Eisberg and Resnick App I, among others):

$\psi_n(u) = H_n(u)e^{-u^2/2}$

Where $H_n(u)$ are Hermite polynomials of order n.

The first few, un-normalised wave functions are

$\psi_0(u) = e^{-u^2/2}$

$\psi_1(u) = ue^{-u^2/2}$

$\psi_2(u) = (2u^2 - 1)e^{-u^2/2}$

The parity is given by the rule $(-1)^n$, where +1 is even and -1 is odd.

The energy eigenvalues can be obtained by substituting into the equation

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+\frac{1}{2}m\omega^2x^2\psi = E\psi$$

For ψ_0, this gives

$$-\frac{\hbar^2}{2m}\left[-\frac{m\omega}{\hbar}+\left(\frac{m\omega x}{\hbar}\right)^2\right]\psi_0+\frac{1}{2}m\omega^2x^2\psi_0 = E\psi_0$$

$$\frac{\frac{\hbar^2}{2m}m\omega}{\hbar}\psi_0 = E_0\,\psi_0 \qquad \rightarrow E_0=\frac{\hbar\omega}{2}$$

This is the lowest energy state (n=0) and again does not have zero energy, due to the zero point motion, there is always some oscillation.

$$u = x\sqrt{\frac{\omega m}{\hbar}} \qquad \frac{du}{dx}=\sqrt{\frac{\omega m}{\hbar}}$$

$$\psi = e^{-\frac{u^2}{2}}$$

$$\frac{d\psi}{dx} = -ue^{-\frac{u^2}{2}}\frac{du}{dx} = -ue^{-\frac{u^2}{2}}\sqrt{\frac{\omega m}{\hbar}}$$

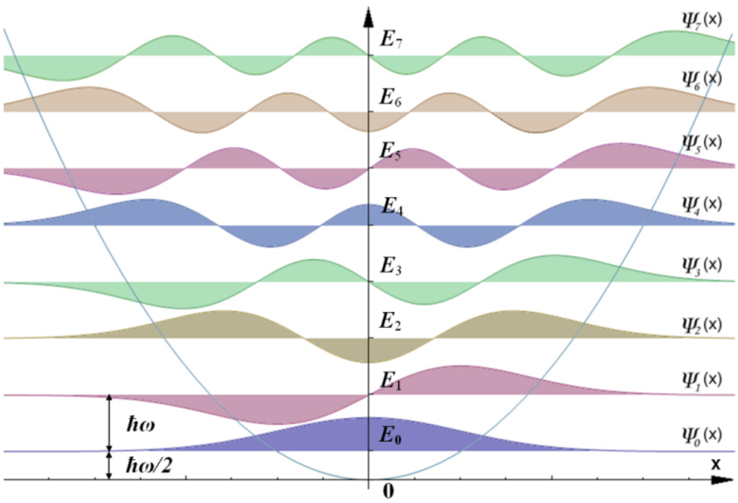
$$\frac{d^2\psi}{dx^2}=\frac{d}{du}\left(-ue^{-\frac{u^2}{2}}\right)\frac{du}{dx}\sqrt{\frac{\omega m}{\hbar}}$$

$$\frac{d^2\psi}{dx^2} = (u^2-1)e^{-\frac{u^2}{2}}\frac{\omega m}{\hbar} = \left[\frac{x^2\omega^2m^2}{\hbar^2}-\frac{\omega m}{\hbar}\right]e^{-u^2/2}$$

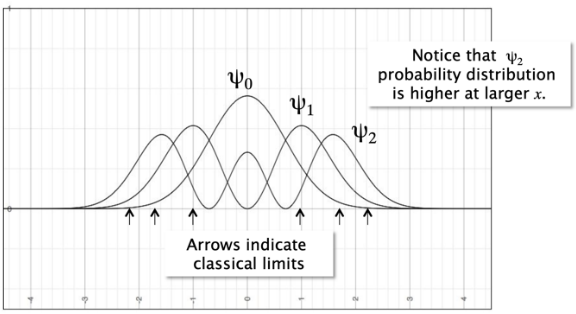
In general, the energies are

$$E_n=\left(n+\frac{1}{2}\right)\hbar\omega$$

- Eigenvalues are linear in n



- As $n \rightarrow \infty$ then oscillations wash out and we reach the behaviour of a classical oscillator (correspondence principle; the correspondence principle states that the behaviour of systems described by the theory of quantum mechanics reproduces classical physics in the limit of large quantum numbers.)



Normalised probability distribution for n=0,1,2